Atrium Healthcare Cost Transparency Database Development

Contents

Part I. Database Initial Study

About the Model Company

Atrium Health is a not-for-profit healthcare system based in Charlotte, North Carolina, with hospitals, clinics, and care centers spread across the Southeast. It offers everything from routine checkups to advanced specialty care, while also supporting medical research and community health programs. Known for its innovation and wide network of providers, Atrium Health works to make quality healthcare more accessible for the communities it serves.

Problems and Constraints

Atrium Health currently faces challenges in its billing process, primarily due to a lack of upfront transparency that leaves patients unaware of treatment costs until after care. Insurance complexity adds to the problem, as co-pays, deductibles, and coverage limits are difficult for patients to understand in advance, often resulting in unexpected charges and dissatisfaction. Cost estimates also vary across departments, creating inconsistencies and eroding trust. The project is constrained by the need to comply with HIPAA regulations, integrate with existing hospital systems, and ensure accurate and timely cost data. Insurance variability further complicates billing, as different providers follow different rules. Additionally, scalability is critical for a large system like Atrium, and successful implementation will require staff training and overcoming resistance to change within a limited time and budget resources.

Project Objectives and Business Needs

This project aims to design and implement a comprehensive relational database system for Atrium Health to significantly enhance cost transparency for medical services by providing patients with clear, pre-service estimates of potential expenses, which addresses the critical business need to eliminate unexpected medical bills and empower patients to make informed healthcare decisions. The new system will also consolidate billing information, efficiently manage appointments, claims, and payments, and enable robust reporting capabilities to improve patient satisfaction, reduce administrative burdens, and ensure Atrium Health remains competitive and compliant in an evolving healthcare landscape.

Scope and Boundaries

Scope

The project will focus on improving Atrium Health's billing process by creating a database in MS Access that provides upfront cost estimates for patients. The system will store standardized treatment, procedure, and medication costs, calculate patient responsibility after insurance coverage, and generate clear billing records. It will also track payments, outstanding balances, and insurance claims to ensure both the patients and hospital staff have accurate and accessible financial information.

Boundaries

The database will not replace Atrium's existing electronic health record or scheduling systems but will function alongside them to support billing transparency. It will not handle complex medical data beyond what is necessary for cost calculation, nor will it serve as a full hospital management system. Insurance rules will be simplified for the project scope, and the system will be designed for demonstration purposes rather than enterprise-scale deployment.

Part II. Database Design

Business Rules

- **Patient ↔ EmergencyContact**: One patient can have many emergency contacts; each emergency contact belongs to exactly one patient.
- **Patient ↔ PreExistingCondition**: One patient can have many pre-existing conditions; each pre-existing condition belongs to exactly one patient.
- **Patient ↔ Treatment**: One patient can have many treatments; each treatment must belong to exactly one patient.
- **Patient ↔ Patient_InsurancePolicy**: One patient can hold many insurance policy records; each policy record belongs to exactly one patient.
- **InsurancePolicy ↔ Patient_InsurancePolicy:** One insurance policy can cover many patient records; each patient record references exactly one insurance policy.
- **Patient ↔ BillingEstimate**: One patient can have many billing estimates; each billing estimate belongs to exactly one patient.
- **Patient ↔ Invoice**: One patient can have many invoices; each invoice belongs to exactly one patient.
- **Patient ↔ Claim**: One patient can file many claims; each claim must belong to exactly one patient.

- **Patient ↔ Medication/Prescription**: One patient can have many prescriptions; each prescription belongs to exactly one patient.

- **Physician ↔ Department**: One department can have many physicians; each physician belongs to exactly one department.
- **Physician ↔ Treatment**: One physician can provide many treatments; each treatment is performed by exactly one physician.
- **Physician ↔ Medication/Prescription**: One physician can issue many prescriptions; each prescription is written by exactly one physician.
- **Physician ↔ BillingEstimate**: One physician can generate many billing estimates; each billing estimate is tied to exactly one physician.

- **Department ↔ Facility/Location**: One facility can have many departments; each department belongs to exactly one facility.
- **Department ↔ Service/Procedure**: One department can offer many services; each service is linked to exactly one department.

- **Facility/Location ↔ Treatment**: One facility can host many treatments; each treatment occurs in exactly one facility.

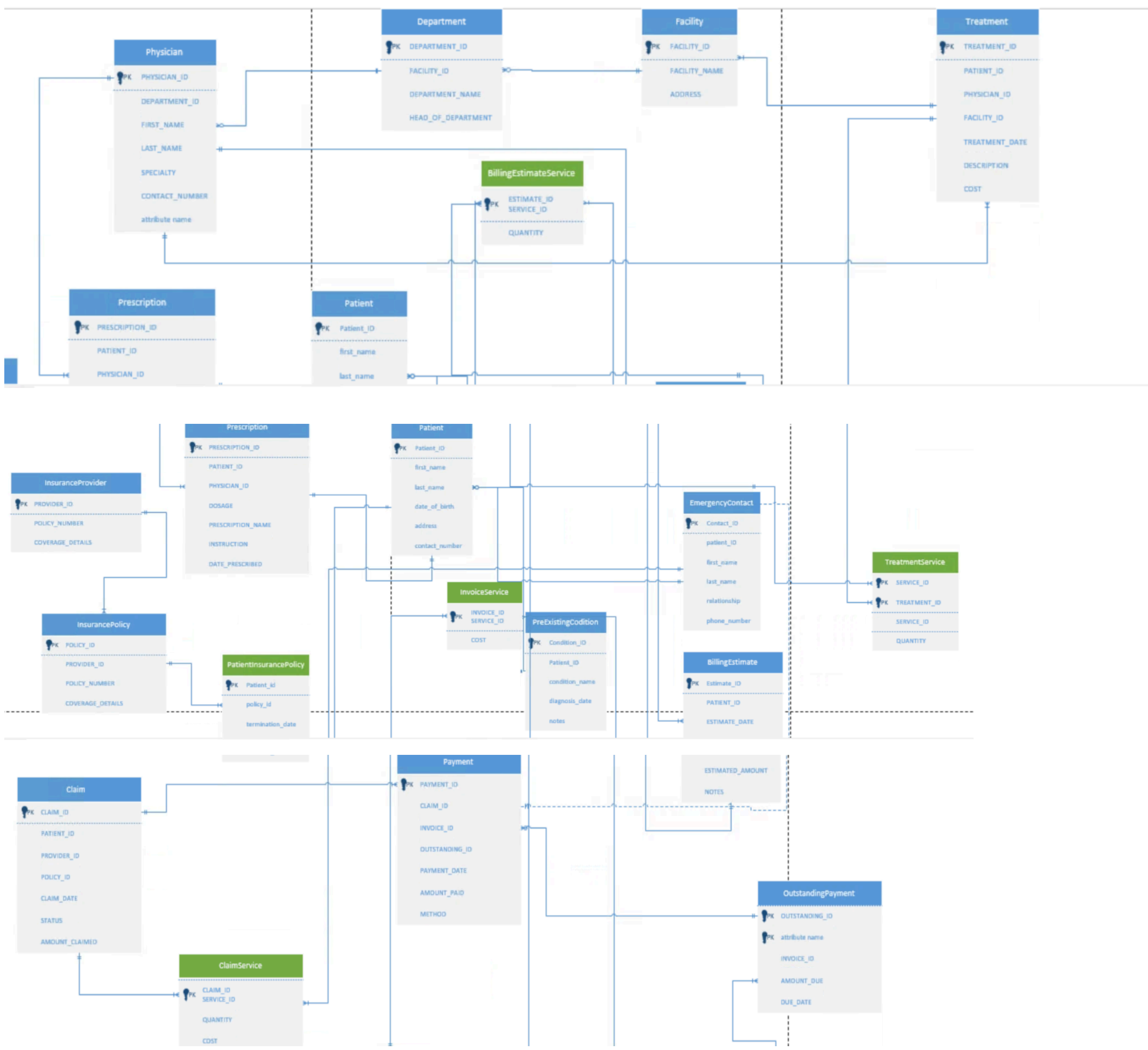
- **Service ↔ Treatment_Service**: One service can be part of many treatment records; each treatment record specifies exactly one service.
- **Treatment ↔ Treatment_Service:** One treatment can consist of many services; each service line belongs to exactly one treatment.
- **Service ↔ BillingEstimate_Service:** One service can appear on many billing estimate records; each billing estimate record specifies exactly one service.
- **BillingEstimate ↔ BillingEstimate_Service:** One billing estimate can include many service items; each service item belongs to exactly one billing estimate.
- **Service ↔ Invoice_Service**: One service can appear on many invoice records; each invoice record specifies exactly one service.
- **Invoice ↔ Invoice_Service:** One invoice can include many service items; each service item belongs to exactly one invoice.
- **Service ↔ Claim_Service**: One service can appear on many claim records; each claim record specifies exactly one service.
- **Claim ↔ Claim_Service:** One claim can include many service items; each service item belongs to exactly one claim.

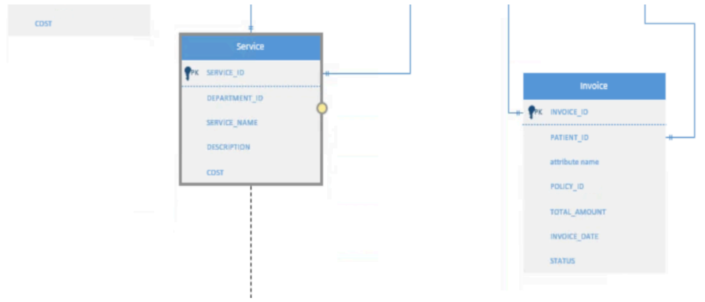
- **InsuranceProvider ↔ InsurancePolicy**: One insurance provider can offer many policies; each policy is offered by exactly one provider.
- **InsuranceProvider ↔ Claim**: One insurance provider can receive many claims; each claim is submitted to exactly one provider.

- **InsurancePolicy ↔ Claim**: One policy can be linked to many claims; each claim must reference exactly one policy.
- **InsurancePolicy ↔ Invoice**: One policy can be linked to many invoices; each invoice can be associated with exactly one policy.

- **Claim ↔ Payment**: One claim can generate many payments; each payment can be linked to exactly one claim.
- **Invoice ↔ Payment**: One invoice can have many payments; each payment is applied to exactly one invoice.
- **Invoice ↔ OutstandingPayment**: One invoice can have many outstanding payment records; each outstanding payment belongs to exactly one invoice.
- **OutstandingPayment ↔ Payment**: One outstanding payment can be reduced by many payments; each payment reduces exactly one outstanding payment.

Normalized ERD





Simplified Data Dictionary

Entity/Table Name	*Attribute/Column Name*	*Key Type*	*Data Type*	*Normal Form*	*Justification / Description*
Patient	***patient_id***	*PK*	*INT*	*3NF*	*Unique identifier for the patient.*
	first_name		*VARCHAR*		*Patient's first name.*
	last_name		*VARCHAR*		*Patient's last name.*
	date_of_birth		*DATE*		*Patient's date of birth.*
	address		*VARCHAR*		*Patient's street address.*
	contact_number		*VARCHAR*		*Patient's phone number.*
---	*---*	*---*	*---*	*---*	*---*
EmergencyContact	***contact_id***	*PK*	*INT*	*3NF*	*Unique identifier for the emergency contact.*
	patient_id	*FK*	*INT*		*References Patient. One patient can have many contacts.*

	first_name		*VARCHAR*		*Contact's first name.*
	last_name		*VARCHAR*		*Contact's last name.*
	relationship		*VARCHAR*		*Relationship to the patient.*
	phone_number		*VARCHAR*		*Contact's phone number.*
---	---	---	---	---	---
PreExistingCondition	***condition_id***	*PK*	*INT*	*3NF*	*Unique identifier for the condition record.*
	patient_id	*FK*	*INT*		*References Patient.*
	condition_name		*VARCHAR*		*Name of the condition.*
	diagnosis_date		*DATE*		*Date the condition was diagnosed.*
	notes		*VARCHAR*		*Additional details.*
---	---	---	---	---	---
Physician	***physician_id***	*PK*	*INT*	*3NF*	*Unique identifier for the physician.*
	department_id	*FK*	*INT*		*References Department.*
	first_name		*VARCHAR*		*Physician's first name.*
	last_name		*VARCHAR*		*Physician's last name.*
	specialty		*VARCHAR*		*Physician's area of expertise.*
	contact_number		*VARCHAR*		*Physician's phone number.*
---	---	---	---	---	---
Department	***department_id***	*PK*	*INT*	*3NF*	*Unique identifier for the department.*

	facility_id	*FK*	*INT*		*References Facility/Location.*
	department_name		*VARCHAR*		*Name of the hospital department.*
	head_of_department		*VARCHAR*		*Name of the person leading the department.*
---	---	---	---	---	---
Facility/Location	***facility_id***	*PK*	*INT*	*3NF*	*Unique identifier for the facility.*
	facility_name		*VARCHAR*		*Name of the facility.*
	address		*VARCHAR*		*Facility's street address.*
---	---	---	---	---	---
Service/Procedure	***service_id***	*PK*	*INT*	*3NF*	*Unique identifier for the medical service.*
	department_id	*FK*	*INT*		*References Department.*
	service_name		*VARCHAR*		*Name of the service.*
	description		*VARCHAR*		*Detailed description of the service.*
	cost		*DECIMAL*		*Standard cost of the service/procedure.*
---	---	---	---	---	---
InsuranceProvider	***provider_id***	*PK*	*INT*	*3NF*	*Unique identifier for the insurance provider company.*
	provider_name		*VARCHAR*		*Name of the insurance company.*
	contact_info		*VARCHAR*		*Contact details for the provider.*
---	---	---	---	---	---
InsurancePolicy	***policy_id***	*PK*	*INT*	*3NF*	*Unique identifier for the insurance policy.*

	provider_id	*FK*	*INT*		*References InsuranceProvider.*
	policy_number		*VARCHAR*		*Unique policy number.*
	coverage_details		*VARCHAR*		*Summary of policy coverage.*
---	---	---	---	---	---
OutstandingPayment	***outstanding_id***	*PK*	*INT*	*3NF*	*Unique identifier for the outstanding payment record.*
	invoice_id	*FK*	*INT*		*References Invoice.*
	amount_due		*DECIMAL*		*The remaining amount that is due.*
	due_date		*DATE*		*The date the outstanding amount is due.*
---	---	---	---	---	---
Treatment	***treatment_id***	*PK*	*INT*	*3NF*	*Unique identifier for the treatment episode.*
	patient_id	*FK*	*INT*		*References Patient.*
	physician_id	*FK*	*INT*		*References Physician.*
	facility_id	*FK*	*INT*		*References Facility/Location.*
	treatment_date		*DATE*		*Date the treatment occurred.*
	description		*VARCHAR*		*Details of the treatment provided.*
	cost	*Derived*	*DECIMAL*		***Derived Attribute: Total cost of treatment. Needed for transaction finality/reporting; usually stored despite being derivable from Treatment_Service.***
---	---	---	---	---	---
Medication/Prescription	***prescription_id***	*PK*	*INT*	*3NF*	*Unique identifier for the prescription.*
	patient_id	*FK*	*INT*		*References Patient.*

	physician_id	*FK*	*INT*		*References Physician.*
	medication_name		*VARCHAR*		*Name of the prescribed medication.*
	dosage		*VARCHAR*		*Amount and frequency.*
	instructions		*VARCHAR*		*Usage instructions.*
	date_prescribed		*DATE*		*Date the prescription was written.*
---	---	---	---	---	---
BillingEstimate	***estimate_id***	*PK*	*INT*	*3NF*	*Unique identifier for the billing estimate.*
	patient_id	*FK*	*INT*		*References Patient.*
	physician_id	*FK*	*INT*		*References Physician.*
	estimate_date		*DATE*		*Date the estimate was generated.*
	estimated_amount	*Derived*	*DECIMAL*		***Derived Attribute: Estimated total cost. Calculated from BillingEstimate_Service.***
	notes		*VARCHAR*		*Notes on the estimate.*
---	---	---	---	---	---
Invoice	***invoice_id***	*PK*	*INT*	*3NF*	*Unique identifier for the invoice.*
	patient_id	*FK*	*INT*		*References Patient.*
	policy_id	*FK*	*INT*		*References InsurancePolicy.*
	invoice_date		*DATE*		*Date the invoice was issued.*
	total_amount	*Derived*	*DECIMAL*		***Derived Attribute: Grand total amount charged. Calculated by summing the cost in Invoice_Service. Needed for transactional integrity.***
	status		*VARCHAR*		*Current status (e.g., 'Pending', 'Paid').*

---	---	---	---	---	---
Claim	***claim_id***	*PK*	*INT*	*3NF*	*Unique identifier for the insurance claim.*
	patient_id	*FK*	*INT*		*References Patient.*
	provider_id	*FK*	*INT*		*References InsuranceProvider.*
	policy_id	*FK*	*INT*		*References InsurancePolicy.*
	claim_date		*DATE*		*Date the claim was filed.*
	status		*VARCHAR*		*Status of the claim.*
	amount_claimed	*Derived*	*DECIMAL*		***Derived Attribute: Total monetary amount claimed. Calculated by summing the cost in Claim_Service. Needed for immediate financial tracking.***
---	---	---	---	---	---
Payment	***payment_id***	*PK*	*INT*	*3NF*	*Unique identifier for the payment transaction.*
	claim_id	*FK*	*INT*		*References Claim (Nullable).*
	invoice_id	*FK*	*INT*		*References Invoice (Nullable).*
	outstanding_id	*FK*	*INT*		*References OutstandingPayment (Nullable).*
	payment_date		*DATE*		*Date the payment was received.*
	amount_paid		*DECIMAL*		*Amount of money paid.*
	method		*VARCHAR*		*Payment method.*
---	---	---	---	---	---
Patient_InsurancePolicy	***patient_id***	*FK, Comp PK*	*INT*	*3NF*	*References Patient. Part of Composite PK.*

	policy_id	*FK, Comp PK*	*INT*		*References InsurancePolicy. Part of Composite PK.*
	effective_date		*DATE*		*Date coverage began for the patient under this policy.*
	termination_date		*DATE*		*Date coverage ended for the patient under this policy.*
---	---	---	---	---	---
Treatment_Service	***treatment_id***	*FK, Comp PK*	*INT*	*3NF*	*References Treatment. Part of Composite PK.*
	service_id	*FK, Comp PK*	*INT*		*References Service/Procedure. Part of Composite PK.*
	quantity		*INT*		*Number of times the service was performed in the treatment.*
---	---	---	---	---	---
BillingEstimate_Service	***estimate_id***	*FK, Comp PK*	*INT*	*3NF*	*References BillingEstimate. Part of Composite PK.*
	service_id	*FK, Comp PK*	*INT*		*References Service/Procedure. Part of Composite PK.*
	quantity		*INT*		*Number of times the service is included in the estimate.*
---	---	---	---	---	---
Invoice_Service	***invoice_id***	*FK, Comp PK*	*INT*	*2NF*	*References Invoice. Part of Composite PK.*
	service_id	*FK, Comp PK*	*INT*		*References Service/Procedure. Part of Composite PK.*
	quantity		*INT*		*Number of times the service is listed on the invoice.*

	cost		*DECIMAL*		***Justification (Not 3NF): cost depends only on service_id (partial dependency, violation of 3NF if cost is supposed to be the standard service cost). However, in billing, the cost must be stored here because the unit price for a service may change over time or be discounted on a specific invoice, thus relying on the full composite key for historical accuracy. This makes it a practical, denormalized design for audit trails.***
---	*---*	*---*	*---*	*---*	*---*
Claim_Service	***claim_id***	*FK, Comp PK*	*INT*	*2NF*	*References Claim. Part of Composite PK.*
	service_id	*FK, Comp PK*	*INT*		*References Service/Procedure. Part of Composite PK.*
	quantity		*INT*		*Number of times the service is included in the claim.*
	cost		*DECIMAL*		***Justification (Not 3NF): Similar to Invoice_Service, cost might violate 3NF if it's the standard price. Practically, the specific cost of the service item at the time of claiming must be stored here for claim processing and audit, making it a necessary deviation from pure 3NF to ensure transactional immutability.***

Notes explaining design decisions and assumptions

- The database is designed mainly in 3rd Normal Form (3NF) to reduce redundancy.
- Some tables use controlled denormalization (like `Invoice_Service` and `Claim_Service`) to keep historical and financial accuracy.
- Primary keys (PK) uniquely identify each record; foreign keys (FK) link related entities.
- Derived fields (like total cost or amount) are stored only when needed for reporting or audit purposes.

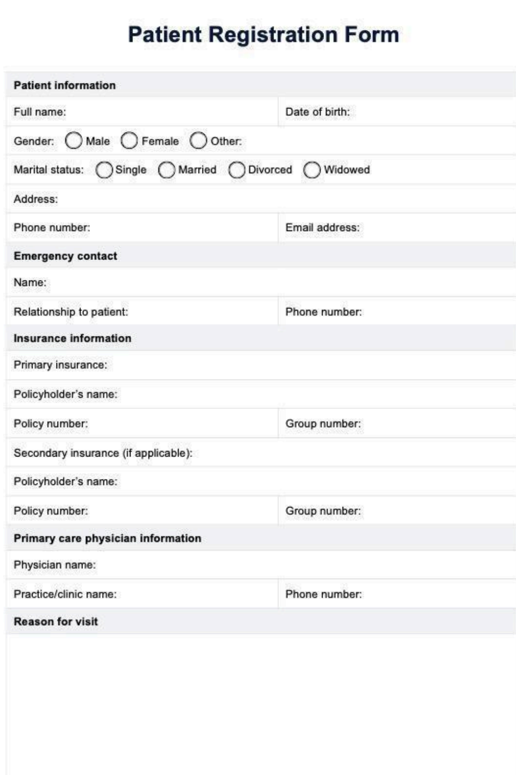
- All foreign keys maintain referential integrity.
- Derived totals (like invoice amounts) are stored for audit consistency.
- Bridge tables handle all many-to-many relationships.

Part III. Implementation and Loading

This part is done in MS Access. Please see the relevant file.

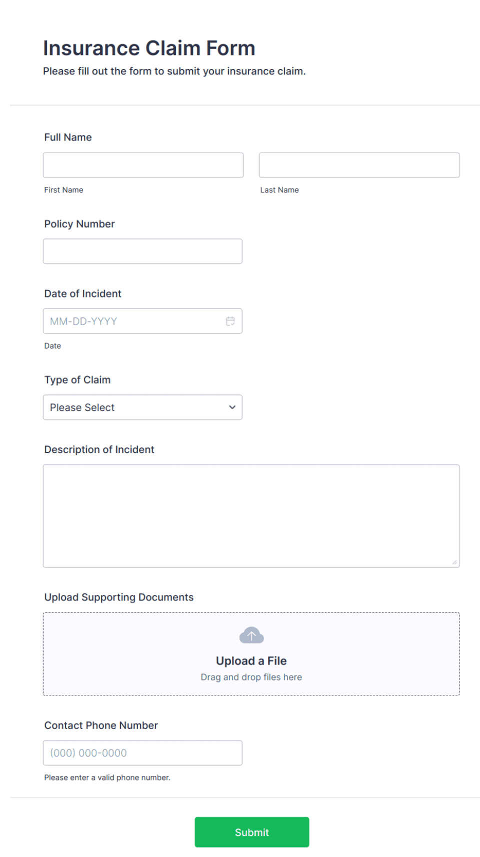
Appendices

Business Forms



Patient Registration Form

Patient information	
Full name:	Date of birth:
Gender: ◯ Male ◯ Female ◯ Other:	
Marital status: ◯ Single ◯ Married ◯ Divorced ◯ Widowed	
Address:	
Phone number:	Email address:
Emergency contact	
Name:	
Relationship to patient:	Phone number:
Insurance information	
Primary insurance:	
Policyholder's name:	
Policy number:	Group number:
Secondary insurance (if applicable):	
Policyholder's name:	
Policy number:	Group number:
Primary care physician information	
Physician name:	
Practice/clinic name:	Phone number:
Reason for visit	



Insurance Claim Form

Please fill out the form to submit your insurance claim.

Full Name

First Name | Last Name

Policy Number

Date of Incident

MM-DD-YYYY

Date

Type of Claim

Please Select

Description of Incident

Upload Supporting Documents

Upload a File

Drag and drop files here

Contact Phone Number

(000) 000-0000

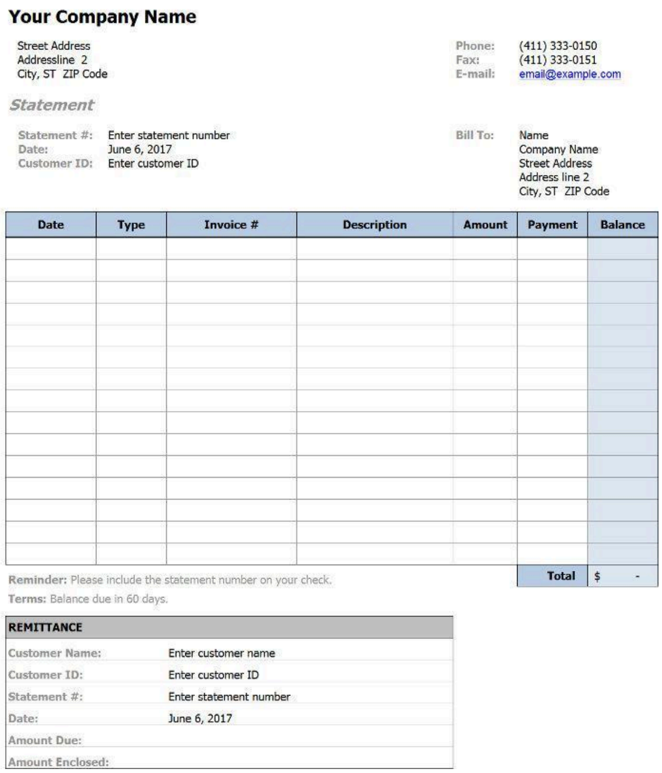
Please enter a valid phone number.

Submit

Above: Patient Registration Form

Above: Insurance Claim Form

Business Reports



Your Company Name

Street Address
Addressline 2
City, ST ZIP Code

Phone: (411) 333-0150
Fax: (411) 333-0151
E-mail: email@example.com

Statement

Statement #: Enter statement number
Date: June 6, 2017
Customer ID: Enter customer ID

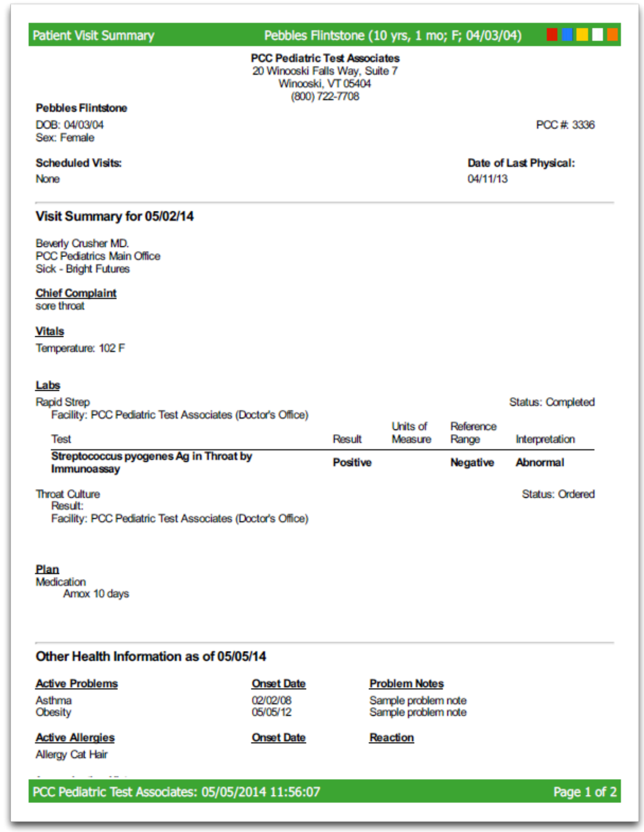
Bill To: Name
Company Name
Street Address
Address line 2
City, ST ZIP Code

Date	Type	Invoice #	Description	Amount	Payment	Balance
					Total	$ -

Reminder: Please include the statement number on your check.
Terms: Balance due in 60 days.

REMITTANCE	
Customer Name:	Enter customer name
Customer ID:	Enter customer ID
Statement #:	Enter statement number
Date:	June 6, 2017
Amount Due:	
Amount Enclosed:	

Above: Billing Summary Report



Patient Visit Summary Pebbles Flintstone (10 yrs, 1 mo; F; 04/03/04)

PCC Pediatric Test Associates
20 Winooski Falls Way, Suite 7
Winooski, VT 05404
(800) 722-7708

Pebbles Flintstone
DOB: 04/03/04
Sex: Female
PCC #: 3336

Scheduled Visits:
None

Date of Last Physical:
04/11/13

Visit Summary for 05/02/14

Beverly Crusher MD.
PCC Pediatrics Main Office
Sick - Bright Futures

Chief Complaint
sore throat

Vitals
Temperature: 102 F

Labs
Rapid Strep Status: Completed
Facility: PCC Pediatric Test Associates (Doctor's Office)

Test	Result	Units of Measure	Reference Range	Interpretation
Streptococcus pyogenes Ag in Throat by Immunoassay	**Positive**		**Negative**	**Abnormal**

Throat Culture Status: Ordered
Result:
Facility: PCC Pediatric Test Associates (Doctor's Office)

Plan
Medication
Amox 10 days

Other Health Information as of 05/05/14

Active Problems	Onset Date	Problem Notes
Asthma	02/02/08	Sample problem note
Obesity	05/05/12	Sample problem note

Active Allergies	Onset Date	Reaction
Allergy Cat Hair		

PCC Pediatric Test Associates: 05/05/2014 11:56:07 Page 1 of 2

Above: Patient Visit Summary Report